

PROGRAM: THREAD CREATION USING PTHREAD

C PROGRAM

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <pthread.h>
```

```
#include <unistd.h>
```

```
void *thread_function(void *arg) {  
    int i;  
    for (i = 1; i <= 5; i++) {  
        printf("Thread Executing : %d\n", i);  
        sleep(1);  
    }  
    pthread_exit(NULL);  
}  
  
int main()  
{  
    pthread_t t1, t2;  
    pthread_create(&t1, NULL, thread_function, NULL);  
    pthread_create(&t2, NULL, thread_function, NULL);  
    pthread_join(t1, NULL);  
    pthread_join(t2, NULL);  
    printf("All Threads Completed\n");  
    return 0;  
}
```

}

SHELL SCRIPT

```
GNU nano 8.7.1
#!/bin/bash

task1()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 1 : $i"
        sleep 1
    done
}

task2()
{
    for i in 1 2 3 4 5
    do
        echo "Thread 2 : $i"
        sleep 1
    done
}

task1 &
task2 &

wait

echo "All Threads Completed"
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano threading.sh

(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x threading.sh

(ganesh@LAPTOP-R3RDH9NG)~$ bash threading.sh
Thread 1 : 1
Thread 2 : 1
Thread 1 : 2
Thread 2 : 2
Thread 2 : 3
Thread 1 : 3
Thread 2 : 4
Thread 1 : 4
Thread 2 : 5
Thread 1 : 5
All Threads Completed
```